Drupal 10 Setup Guide on Ubuntu 24.02 LTS

1. Initial Setup

Install Ubuntu 24.02 LTS on VirtualBox

On Ubuntu 24.02 LTS, ensure it is updated:

```
sudo apt update && sudo apt upgrade -y
```

2. Install LAMP Stack (Linux, Apache, MySQL, PHP)

Install Apache

Apache is the web server that will serve your Drupal site.

```
sudo apt install apache2 -y
```

Enable and start Apache:

```
sudo systemctl enable apache2
```

```
sudo systemctl start apache2
```

Install MySQL

MySQL will be the database management system.

```
sudo apt install mysql-server -y
```

Secure the MySQL installation:

```
sudo mysql_secure_installation
```

Follow the prompts to secure your MySQL installation (set a root password, remove anonymous users, disallow root login remotely, remove test database, and reload privilege tables).

Install PHP

PHP is the scripting language required by Drupal.

```
sudo apt install php libapache2-mod-php php-mysql php-gd php-xml php-mbstring
php-curl php-zip php-json -y
```

Restart Apache to apply the changes:

```
sudo systemctl restart apache2
```

3. Install Drupal

Install Composer

Composer is a dependency manager for PHP.

```
sudo apt install composer -y
```

Set Up MySQL Database for Drupal

Log in to MySQL as the root user:

```
sudo mysql -u root -p
```

Create a database and user for Drupal:

```
CREATE DATABASE drupaldb;

CREATE USER 'drupaluser'@'localhost' IDENTIFIED BY 'password';

GRANT ALL PRIVILEGES ON drupaldb.* TO 'drupaluser'@'localhost';

FLUSH PRIVILEGES;

EXIT;
```

Replace password with a secure password.

Download and Install Drupal

Navigate to the web root directory:

```
cd /var/www/html
```

Use Composer to create a new Drupal project:

```
sudo composer create-project drupal/recommended-project drupal
```

Set the correct permissions:

```
sudo chown -R www-data:www-data /var/www/html/drupal

sudo chmod -R 755 /var/www/html/drupal
```

Create a virtual host configuration for Apache:

```
sudo nano /etc/apache2/sites-available/drupal.conf
```

Add the following configuration:

```
<VirtualHost *:80>
ServerAdmin webmaster@localhost
DocumentRoot /var/www/html/drupal/web
ServerName drupal.local
<Directory /var/www/html/drupal/web>
Options Indexes FollowSymLinks
AllowOverride All
Require all granted
</Directory>
ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

Enable the site and rewrite module:

```
sudo a2ensite drupal.conf
```

```
sudo a2enmod rewrite
```

```
sudo systemctl restart apache2
```

Edit Hosts File

Add an entry for drupal.local:

```
sudo nano /etc/hosts
```

Add the following line:

```
127.0.0.1 drupal.local
```

4. Complete Drupal Setup Through Browser

Open your web browser and go to http://drupal.local. Follow the installation steps provided by the Drupal setup wizard:

1. Select an installation profile - Standard is recommended.

2. Set up database - Enter the database name (drupaldb), username (drupaluser), and password you created earlier.

3. Configure site - Enter site information such as site name, site email address, and admin account details.

5. Post-Installation

Once the setup is complete, you should secure your Drupal installation:

Set File Permissions

```
sudo chown -R www-data:www-data /var/www/html/drupal

sudo chmod -R 755 /var/www/html/drupal
```

6. Additional Configuration

Enable Clean URLs

Drupal often requires Clean URLs to be enabled. Ensure .htaccess is working:

```
sudo nano /var/www/html/drupal/web/.htaccess
```

Uncomment or add the following line if necessary:

```
RewriteEngine on
```

Verify the mod_rewrite module is enabled:

```
sudo a2enmod rewrite

sudo systemctl restart apache2
```